MIL-C-63177A (AR)
16 May 1978
SUPERSEDING
MIL-C-63177
29 November 1976

MILITARY SPECIFICATION

CONTROL UNIT: LANCE AK (M238E1)

This specification is approved for use by U.S. Army Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of subassembly, entitled Control Unit, which is used in the Lance warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-48305	- Squib Switches.
MIL-G-48529	- Rate Gyroswitch.
MIL-P-63176	- Printed Wiring Board.
MIL-D-48522	- Delay Unit.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

FSC NUOR

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

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MIL-STD-1460 - Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS (see 6.5)

U.S. ARMY RESEARCH AND DEVELOPMENT COMMAND
PRODUCT AND PACKAGING DRAWINGS

9236970	- Plate Mounting (Casting).
9236980	- Mounting Plate Assembly.
9236990	- Cover.
9317506	- Control Unit.
ET-9317056	- Equipment Tabulation.
9237017	- Plate, Connector.
9237019	- Plate, Connector Assembly.
9317505	- Terminal Board and Harness Assembly.
9317502	- Printed Wiring Board Assembly.
9317504	- Terminal Board, Component in Jumper Installation.
9237608	- Energy Transfer Module (ETM).

PUBLICATIONS (see 6.5)

U.S. ARMY RESEARCH AND DEVELOPMENT COMMAND

QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The Control Unit shall comply with all requirements specified on the drawing (dwg) 9317506, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

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3.4 Serialization. A serial number shall be assigned to the item. Serial numbers required on these items shall be obtained by means of a letter addressed to: Commander, ARRADCOM, Dover, New Jersey 07801, ATTENTION: DRDAR-QAN-R. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial numbers required

3.5 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

3.6 Components. For components to be used in the Control Unit, objective evidence (see 6.2) indicating that the following components conform to their respective specifications and/or drawings shall be made available to the Government representative:

a. M46 Rate Gyroswitch - MIL-G-48529(PA).
b. Squib Switches - MIL-S-48305(PA).
c. Printed Wiring Board - MIL-P-63176(PA).
d. Delay-Unit - MIL-D-48522(PA).
e. Plate Mounting (Casting) - Drawing 9236970.
f. Energy Transfer Module (ETM) - MIL-M-63174.

3.7 Circuit resistance - PWB #1 Assembly dwg 9317502 (Before Conformal Coating (BCC) and After Conformal Coating (ACC)): The resistance of each circuit shall not exceed the maximum value listed in Table I or as shown otherwise when tested as specified in 4.4.1.1.

Table I

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)
J19-1W	J19-4V	0.3	060001
J19-1V	J19-4W	0.3	060002
J19-1U	J19-1R	0.3	060003
J19-1T	J19-1S	0.3	060004
J19-1P	J19-4N	0.3	060005
J19-1P	J19-4M	0.3	060006
J19-1L	J19-4P	1.5-2.5	060007
J19-1L	J19-4R	24-32	060008
J19-2W	J19-4L	0.3	060009
J19-2W	J19-2A	12-16	060010
J19-2V	J19-3V	0.3	060011

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Table I (Cont'd)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)
J19-2U	J19-2T	1.5-2.5	060012
J19-2T	J19-2S	10-14	060013
J19-2S	J19-4R	0.3	060014
J19-2R	J19-5W	0.3	060015
J19-2N	J19-2M	0.3	060016
J19-2P	177	0.3 (BCC only)	060017
J19-2L	181	0.3 (BCC only)	060018
J19-5U	J19-5T	1.5-2.5	060019
J19-3W	J19-4E	0.3	060020
J19-5R	159	0.3 (BCC)	060021
J19-3U	J19-3S	0.3	060022
J19-3U	J19-4R	12-16	060023
J19-3T	J19-3R	0.3	060024
J19-1E	J19-2J	1.5-2.5	060025
J19-1K	126	0.3 (BCC only)	060026
J19-1J	122	0.3 (BCC only)	060027
J19-1H	121	0.3 (BCC only)	060028
J19-1G	120	0.3 (BCC only)	060029
J19-1F	119	0.3 (BCC only)	060030
J19-1E	J19-1D	1.5-2.5	060031
J19-1E	J19-1B	12-16	060032
J19-1E	J19-2K	12-16	060033
J19-1E	J19-2H	2.5-4.0	060034
J19-1C	124	0.3 (BCC only)	060035
J19-1B	J19-3P	0.3	060036
J19-1B	J19-3N	0.3	060037
J19-5S	149	0.3 (BCC)	060038
J19-1A	130	0.3 (BCC only)	060039
J19-2J	J19-2H	1.5-2.5	060040
J19-2K	J19-1B	0.3	060041
J19-2G	132	0.3 (BCC only)	060042
J19-2F	133	0.3 (BCC only)	060043
J19-2E	134	0.3 (BCC only)	060044
J19-2D	135	0.3 (BCC only)	060045
J19-2C	131	0.3 (BCC only)	060046
J19-2B	J19-3G	0.3	060047
J19-3J	J19-3F	0.3	060048
J19-3J	J19-3E	12-16	060049
J19-5L	J19-5M	0.3	060050
J19-3K	J19-3H	0.3	060051
J19-3C	J19-4G	0.3	060052
J19-3B	J19-4J	0.3	060053
J19-3A	J19-4K	0.3	060054
J19-3B	J19-4H	0.75-2	060055
J19-4U	J19-4S	24-32	060056

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Table I (Cont'd)

Terminals From	To	Resistance (ohms) (max)	Test Code No. (see 3.5)
J19-4U	J19-4R	0.3	060057
J19-4T	J19-4S	1.5-2.5	060058
J19-4R	J19-4H	12-16	060059
J19-4D	J19-4C	1.5-2.5	060060
J19-4B	J19-4C	0.3	060061
J19-4B	J19-4A	1.5-2.5	060062
J19-5V	J19-4R	0.3	060063
J19-5V	J19-5U	10-14	060064
J19-1F	J19-3V	4-8	060065
J19-5K	191	0.3 (BCC)	060066
J19-5J	192	0.3 (BCC)	060067
J19-4P	145	0.3 (BCC)	060068
J19-4T	146	0.3 (BCC)	060069
J19-2U	172	0.3 (BCC)	060070
J19-5T	183	0.3 (BCC)	060071
J19-40	104	0.3 (BCC)	060072
J19-4A	105	0.3 (BCC)	060073
J19-1D	J19-2J	0.3	060074
J19-3V	J19-2G	4-8	060075
J19-4B	#72 hole (9317501 sheet 2)	4-8 (BCC)	060076

3.8 Insulation resistance - printed wiring board assembly dwg 9317502. The insulation resistance between each circuit and all other circuits, as indicated in Table II, shall not be less than 100 megohms when tested as specified in 4.4.2.1.

Table II

From	To	From	To
J19-1A	All others & shell	J19-2L	All others & shell
J19-1B	"	J19-2M	"
J19-1C	"	J19-2N	"
J19-1G	"	J19-2P	"
J19-1H	"	J19-2R	"
J19-1J	"	J19-3A	"
J19-1K	"	J19-3C	"
J19-1L	"	J19-3H	"
J19-1P	"	J19-3K	"
J19-1R	"	J19-3R	"
J19-1T	"	J19-3W	"
J19-1U	"	J19-4A	"
J19-1V	"	J19-5J	"
J19-1W	"	J19-5K	"

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Table II (Cont'd)

From	To	From	To
J19-2A	All others & shell	J19-5L	All others & shell
J19-2B	"	J19-5R	"
J19-2C	"	J19-5S	"
J19-2D	"	J19-1S	"
J19-2E	"		
J19-2F	"		

The following list of unused pins may be connected and tested as one terminal: 1N, 1M, 3M, 3L, 5P, 5N, 3D, 4F, 5F, 5L 5D, 5C, 5B, 5A, 5G, 5H.

3.8.1 One megohm insulation resistance - printed wiring board assembly dwg 9317502. The insulation resistance between each circuit (including those of Table II) as indicated in Table IIa shall not be less than one megohm when tested as specified in 4.4.2.2.

Table IIa

From	To	From	To
J19-1L	All others & shell	J19-1B	All others & shells
J19-2A	"	J19-1F	"
J19-4A	"	#166 hole (9317501 sheet 2)	

3.9 Circuit resistance - term, bd harness assembly dwg 9317505 (BCC and ACC). The resistance of each circuit shall not exceed the maximum value listed in Table III, or as shown otherwise when tested as specified in 3.9.1, 3.9.2 and 4.4.1.1.

3.9.1 Relay coil test voltage polarity. Certain test codes listed in Table III require that a relay be energized with 30 + 2v to perform the test. When a relay "ON" requirement is listed for a test code, the relay shall be energized by a floating deck power supply connected as follows:

Relay Desig	Polarity +	Polarity -
K1A	P19-5S	P19-2A
K1B	P19-5R	P19-2A
K2A	1J2-a	1J2-D
K2B	1J2-$\overline{S}$	1J2-U

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3.9.2 Test terminal polarity. Certain test codes require a voltage of proper polarity to be used to obtain the test result indicated in Table III. Test voltage polarity shall be used, as indicated by symbols appearing in Table III, as (+) or (-) following the terminal to which they apply.

Table III

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)	Req'd
P19-1W	P19-4M	0.3	080001	
P19-1W	P8-H	0.3	080002	
P19-1W	P8-K	0.3	080003	
P19-1W	P8-J	0.3	080004	
P19-1W	P8-L	0.3	080005	
P19-1V	P19-4N	0.3	080006	
P19-1V	P9-H	0.3	080007	
P19-1V	P9-J	0.3	080008	
P19-1V	P9-K	0.3	080009	
P19-1V	P9-L	0.3	080010	
P19-1U	P7-J	0.3	080011	
P19-1U	P7-H	0.3	080012	
P19-1T	P6-J	0.3	080013	
P19-1T	P6-H	0.3	080014	
P19-1S	J3-c	0.3	080015	
P19-1S	CR203-A	0.3 (BCC)	080016	
P19-1R	J3-w	0.3	080017	
P19-1R	CR202-A	0.3 (BCC)	080018	
P19-1P	J3-L	21-33	080019	
P19-1L	P5-29	0.3	080020	
P19-1J	1J1-D	0.3	080021	
P19-1J	1J1-E	0.3	080022	
P19-1J	CR-410-C	0.3 (BCC)	080023	
P19-2V	1J12-G	0.3	080024	
P19-2V	1J12-H	0.3	080025	
P19-2V	1J12-J	0.3	080026	
P19-2V	1J12-X	0.3	080027	
P19-2V	P5-31	0.3	080028	
P19-1H	1J1-M	0.3	080029	
P19-1H	1J13-62	0.3	080030	
P19-1H	P19-2E	0.3	080031	
P19-1H	1J1-L	0.3	080032	
P19-1G	1J1-K	0.3	080033	
P19-1G	1J1-b	0.3	080034	
P19-1G	P19-2F	0.3	080035	
P19-1G	1J13-63	0.3	080036	
P19-1F	P4-Z	0.3	080037	
P19-1E	P5-23	0.3	080038	

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Table III (Cont'd)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)	Req'd
P19-1D	P5-51	0.3	080039	
P19-1C	J3-v	0.3	080040	
P19-1B	P4-$\overline{A}$A	0.3	080041	
P19-1B	P4-BB	0.3	080042	
P19-1B	P19-2K	0.3	080043	
P19-1A	J3-d	0.3	080044	
P19-2W	J15-$\overline{M}$	0.3	080045	
P19-2U	CR205-C	0.3 (BCC)	080046	
P19-2U	P5-7	0.3	080047	
P19-2U	P8-P	0.3	080048	
P19-2U	P9-U	0.3	080049	
P19-2U	1J13-9	0.3	080050	
P19-2T	P9-M	0.3	080051	
P19-2T	P8-N	0.3	080052	
P19-2T	P5-5	0.3	080053	
P19-2T	P19-2S	5-7	080054	
P19-2P	1J1-C	0.3	080055	
P19-2P	1J1-S	0.3	080056	
P19-2P	1J1-V	0.3	080057	
P19-2P	1J1-d	0.3	080058	
P19-2P	1J2-$\overline{L}$	0.3	080059	
P19-2P	1J2-V	0.3	080060	
P19-2P	P4-V	0.3	080061	
P19-2P	P4-c	0.3	080062	
P19-2P	P19-$\overline{2}$L	0.3	080063	
P19-2N	P8-E	0.3	080064	
P19-2N	P8-B	0.3	080065	
P19-2N	P8-C	0.3	080066	
P19-2N	P8-D	0.3	080067	
P19-2N	P9-B	0.3	080068	
P19-2N	P9-C	0.3	080069	
P19-2N	P9-D	0.3	080070	
P19-2N	P9-E	0.3	080071	
P19-2N	1J13-25	0.3	080072	
P19-2N	P19-2M	0.3	080073	
P19-2N	J3-E	0.3	080074	
P19-2J	P5-45	0.3	080075	
P19-2J	1J13-15	0.3	080076	
P19-2H	P5-22	0.3	080077	
P19-2G	P4-b	0.3	080078	
P19-2B	1J2-$\overline{M}$	0.3	080079	
P19-2D	1J1-W	0.3	080080	
P19-2D	1J1-X	0.3	080081	
P19-2D	CR412-C	0.3 (BCC)	080082	
P19-2A	P6-B	0.3	080083	
P19-2A	P6-C	0.3	080084	

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Table III (Cont'd)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)	Req'd
P19-2A	P6-E	0.3	080085	
P19-2A	P6-G	0.3	080086	
P19-2A	P7-B	0.3	080087	
P19-2A	P7-C	0.3	080088	
P19-2A	P7-E	0.3	080089	
P19-2A	P5-9	0.3	080090	
P19-2A	P5-16	0.3	080091	
P19-2A	P5-47	0.3	080092	
P19-2A	P5-48	0.3	080093	
P19-2A	P4-M	0.3	080094	
P19-2A	P4-N	0.3	080095	
P19-2A	1J13-24	0.3	080096	
P19-2A	P19-3E	0.3	080097	
P19-2A	TD201-1	0.3 (BCC)	080098	
P19-2A	E-202	0.3 (BCC)	080099	
P19-2A	P7-D	0.3	080100	
P19-2A	P7-F	0.3	080101	
P19-2A	P6-F	0.3	080102	
P19-2A	P6-D	0.3	080103	
P19-2A	P7-A	14-16	080104	
P19-2A	P7-L	14-16	080105	
P7-T	P7-K	0.3	080106	
P6-T	P6-K	0.3	080107	
P19-2A	P6-A	14-16	080108	
P19-2A	P6-L	14-16	080109	
P19-3W	1J13-44	0.3	080110	
P19-3V	1J12-F	0.3	080111	
P19-3V	1J12-Y	0.3	080112	
P19-3V	1J12-Z	0.3	080113	
P19-3V	1J12-f	0.3	080114	
P19-3V	P5-30	0.3	080115	
P19-5L	1J2-F	0.3	080116	
P19-5M	1J2-W	0.3	080117	
P19-3A	1J16-S	0.3	080118	
P19-3A	1J17-S	0.3	080119	
P19-3U	1J2-E	0.3	080120	
P19-3U	1J13-6	0.3	080121	
P19-3T	P10-V	0.3	080122	
P19-3S	1J2-Y	0.3	080123	
P19-3R	P10-W	0.3	080124	
P19-3R	1J13-5	0.3	080125	
P19-3P	J3-HH	0.3	080126	
P19-2A	P10-D	0.3	080127	

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Table III (Cont'd)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)	Req'd
P19-2A	P10-N	0.3	080128	
P19-2A	P7-G	0.3	080129	
P19-3N	J3-u	0.3	080130	
P19-3K	1J2-A	0.3	080131	
P19-3J	J3-M	0.3	080132	
P19-3J	P19-3F	0.3	080133	
P19-3J	1J13-3	0.3	080134	
P19-3J	CR202-C	0.3 (BCC)	080135	
P19-3H	1J2-J	0.3	080136	
P19-3G	1J2-G	0.3	080137	
P19-3C	P5-25	0.3	080138	
P19-3B	P19-4J	0.3	080139	
P19-3B	P5-50	0.3	080140	
P19-3B	1J13-27	0.3	080141	
P19-3A	P19-4K	0.3	080142	
P19-4W	CR208-A	0.3 (BCC)	080143	
P19-4W	CR207-A	0.3 (BCC)	080144	
P19-4V	CR209-A	0.3 (BCC)	080145	
P19-4V	CR210-A	0.3 (BCC)	080146	
P19-4U	P4-X	0.3	080147	
P19-4U	P19-4R	0.3	080148	
P19-4U	P10-Y	0.3	080149	
P19-4U	TD201-8	0.3 (BCC)	080150	
P19-4U	P4-Y	0.3	080151	
P19-4U	P10-Z	0.3	080152	
P19-4U	P5-12	8-12	080153	
P19-4U	P5-11	8-12	080154	
P19-4T	P19-4P	0.3	080155	
P19-4T	P5-20	0.3	080156	
P19-4T	P5-21	0.3	080157	
P19-4T	1J13-8	0.3	080158	
P19-4T	CR205-A	0.3 (BCC)	080159	
P19-4S	P5-28	0.3	080160	
P19-4H	P5-49	0.3	080161	
P19-4G	P5-4	0.3	080162	
P19-4L	J15-L	0.3	080163	
P19-4E	J3-J	0.3	080164	
P19-4E	1J13-49	0.3	080165	
P19-4D	P5-52	0.3	080166	
P19-4C	P5-18	0.3	080167	
P19-4B	P5-19	0.3	080168	
P19-4B	1J13-14	0.3	080169	
P19-4A	P5-46	0.3	080170	
P19-5V	P19-5U	5.4-6.9	080171	

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Table III (Cont'd)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)	Req'd
P19-5V	P5-8	5.4-6.9	080172	
P19-5V	P9-N	5.4-6.9	080173	
P19-5V	P8-M	5.4-6.9	080174	
P19-5T	P5-6	0.3	080175	
P19-5T	P9-P	0.3	080176	
P19-5T	P8-U	0.3	080177	
P19-5T	1J13-10	0.3	080178	
P19-5S	J15-R	0.3	080179	
P19-5S	1J13-35	0.3	080180	
P19-5R	J15-S	0.3	080181	
P19-5R	1J13-39	0.3	080182	
P4-A	1J2-H	0.3	080183	
P4-A	1J13-30	0.3	080184	
P4-B	1J13-29	0.3	080185	
P4-C	P4-D	0.3	080186	
P4-C	J3-R	0.3	080187	
P4-S	P5-26	0.3	080188	
P4-G	P5-27	0.3	080189	
P4-G	P5-33	0.3	080190	
P4-E	1J12-R	0.3	080191	
P4-F	1J12-e	0.3	080192	
P4-E	1J13-22	0.3	080193	
P4-G	1J13-57	0.3	080194	
P4-G	P4-H	0.3	080195	
P4-J	P4-K	0.3	080196	
P4-S	1J13-60	0.3	080197	
P4-T	P4-U	0.3	080198	
P4-W	1J13-28	0.3	080199	
P4-a	P5-24	0.3	080200	
P4-a	1J13-31	0.3	080201	
P4-CC	1J13-18	0.3	080202	
P4-CC	1J12-c	0.3	080203	
P4-CC	P4-DD	0.3	080204	
P4-CC	1J12-N	0.3	080205	
P5-35	P5-43	0.3	080206	
P5-35	P5-36	0.3	080207	
P5-39	P6-S	0.3	080208	
P5-39	1J13-33	0.3	080209	
P5-15	P5-14	540-660	080210	
P6-M	P6-N	0.3	080211	
P6-M	P7-M	24-36	080212	
P6-M	P7-N	24-36	080213	
P6-U	P7-U	0.3	080214	
P6-U	1J2-D	0.3	080215	
P6-U	1J2-X	0.3	080216	

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Table III (Cont'd)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)	Req'd
P6-U	1J13-50	0.3	080217	
P6-U	K2A-20	0.3 (BCC)	080218	
P6-R	P7-S	0.3	080219	
P6-R	1J13-34	0.3	080220	
P6-P	P7-P	0.3	080221	
P6-P	1J2-K	0.3	080222	
P6-P	1J2-U	0.3	080223	
P6-P	K2B-20	0.3 (BCC)	080224	
P7-R	P8-R	0.3	080225	
P7-R	1J13-36	0.3	080226	
P8-S	P9-R	0.3	080227	
P8-S	1J13-37	0.3	080228	
P9-S	P10-a	0.3	080229	
P9-S	1J13-38	0.3	080230	
P10-E	J15-H	0.3	080231	
P10-E	1J13-40	0.3	080232	
J15-D	J15-E	0.3	080233	
J15-D	1J13-48	0.3	080234	
J15-G	1J13-44	0.3	080235	
J15-N	J15-P	0.3	080236	
J15-N	1J13-51	0.3	080237	
1J1-A	1J1-U	0.3	080238	
1J1-B	1J13-47	0.3	080239	
1J1-F	1J1-Z	0.3	080240	
1J1-F	J3-j	0.3	080241	
1J1-G	1J1-Y	0.3	080242	
1J1-G	J3-MM	0.3	080243	
1J1-H	1J2-B	0.3	080244	
1J1-H	1J13-42	0.3	080245	
1J1-P	J3-W	0.3	080246	
1J1-R	J3-S	0.3	080247	
1J1-T	1J1-e	0.3	080248	
1J2-N	1J2-P	0.3	080249	
1J2-N	J3-D	0.3	080250	
1J2-S	1J13-46	0.3	080251	
1J2-S	K2B-19	0.3 (BCC)	080252	
1J2-a	1J13-43	0.3	080253	
1J2-a	K2A-19	0.3 (BCC)	080254	
P19-5K	P4-C	0.3	080255	
P5-14	1J13-12	0.3	080256	
P10-C	1J13-1	0.3	080257	
P10-J	1J13-58	0.3	080258	
P10-M	1J13-61	0.3	080259	
P10-R	1J13-32	0.3	080260	
P10-U	1J13-56	0.3	080261	
P10-b	1J13-64	0.3	080262	

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Table III (Cont'd)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)	Req'd
P19-2A	P5-35	0.3	080263	
P19-2A	P5-36	0.3	080264	
P19-2A	P5-43	0.3	080265	
P5-26	E-331	0.3 (BCC)	080266	
P5-33	E-305	0.3 (BCC)	080267	
P5-27	E-305	0.3 (BCC)	080268	
P5-47	E-322	0.3 (BCC)	080269	
P5-48	E-314	0.3 (BCC)	080270	
P4-E	1J1-e	0.3	080271	
P4-E	1J1-$\overline{T}$	0.3	080272	
P4-F	1J1-A	0.3	080273	
P4-F	1J1-U	0.3	080274	
1J16-C	1J12-f	80-120	080275	
1J17-C	1J12-$\overline{X}$	80-120	080276	
CR-403-C	CR-410-A	0.3 (BCC)	080277	
CR-406-C	CR-412-A	0.3 (BCC)	080278	
CR-403-G	1J16-C	0.3 (BCC)	080279	
CR-406-G	1J17-C	0.3 (BCC)	080280	
J3-BB	ETM 2-IN	0.3 (BCC)	080281	
J3-z	ETM 1-IN	0.3 (BCC)	080282	
ETM 1-OUT	CR-403-A	0.3 (BCC)	080283	
ETM 1-OUT	1J13-59	0.3 (BCC)	080284	
ETM 2-OUT	CR-406-A	0.3 (BCC)	080285	
ETM 2-OUT	1J13-65	0.3 (BCC)	080286	
ETM 1-C	1J12-F	0.3 (BCC)	080287	
ETM 2-C	1J12-X	0.3 (BCC)	080288	
1J13-13	J3-p	0.3	080289	
1J13-13	J3-H	0.3	080290	
J3-F	J3-i	0.3	080291	
J3-U	J3-$\overline{LL}$	0.3	080292	
J3-Y	J3-JJ	0.3	080293	
J3-a	J3-NN	0.3	080294	
J3-$\overline{q}$	J3-KK	0.3	080295	
J3-k	J3-FF	0.3	080296	
J3-$\overline{CC}$	P10-1B	0.3	080297	
J3-EE	P19-1B	0.3	080298	
1J2-C	1J2-R	0.3	080299	
P5-15	1J13-11	0.3	080300	
P5-15	CR211-A	0.3 (BCC)	080301	
P5-15	E203	270-330 (BCC)	080302	
P5-15	1J13-12	540-660	080303	
P5-15	CR212-A	540-660 (BCC)	080304	
J15-A	CR211-C	0.3 (BCC)	080305	
J15-A	CR212-C	0.3 (BCC)	080306	
J15-A	1J13-7	0.3	080307	

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Table III (Cont'd)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)	Req'd
J15-A	J15-B	0.3	080308	
1J2-c	P5-41	0.3	080309	
1J2-b	P5-40	0.3	080310	
P19-2R	J3-n	0.3	080311	K1A-ON
P19-1P	E435	0.3 (BCC)	080312	
E436	J3-L	0.3 (BCC)	080313	
P19-4W(+)	J3-p (-)	50-74	080314	
P19-4V(+)	J3-H (-)	50-74	080315	
J3-s	P19-5W	0.3	080316	K1B-ON
P19-3V	E410	0.3 (BCC)	080317	
CR403C(+)	1J1-D (-)	50-74 (BCC)	080318	
P19-2V	E412	0.3 (BCC)	080319	
CR406C(+)	P19-2D (-)	50-74 (BCC)	080320	
J15-N	J15-P	0.3	080321	
P6-K (+)	P4-T (-)	50-74	080322	
P7-T (+)	P4-U (-)	50-74	080323	
P19-2S	E442	0.3 (BCC)	080324	
P5-6	CR206C	0.3 (BCC)	080325	
P19-4T	CR206A	0.3 (BCC)	080326	
P19-4T(+)	P5-6 (-)	50-74	080327	
P19-4T(+)	P19-2U (-)	50-74	080328	
P5-8	P9-N	0.3	080329	
P19-5V	E434	0.3 (BCC)	080330	
1J2-S (+)	P7-P (-)	180-220	080331	
P7-P (+)	1J2-S (-)	50-80	080332	
P6-U (+)	1J2-a (-)	50-80	080333	
1J2-a(+)	P6-U (-)	180-220	080334	
1J13-55	1J13-54	1 meg (min)	080335	K2B-ON
1J13-54	1J13-53	1 meg (min)	080336	K2A-ON
1J13-53	1J13-52	1 meg (min)	080337	K1B-ON
1J13-52	1J13-45	1 meg (min)	080338	K1A-ON
P19-1R(+)	J3-M (-)	50-74	080339	
J3-c (+)	1J13-3 (-)	50-74	080340	
J3-C	E444	0.3 (BCC)	080341	K2A-ON
J3-C	E444	0.3 (BCC)	080342	K2B-ON
P6-M	E444	12-17 (BCC)	080343	
P7-M	E444	12-17 (BCC)	080344	
P5-15(+)	J15-A (-)	50-74	080345	
E218	P5-14	0.3 (BCC)	080346	
E218 (+)	J15-B (-)	50-74 (BCC)	080347	
P7-L (+)	P4-K (-)	50-74	080348	

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Table III (Cont'd)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)	Req'd
P7-L	P7-A	0.3	080349	
P6-A (+)	P4-K (-)	50-74	080350	
P6-A	P6-L	0.3	080351	
P19-2A(+)	P19-5S (-)	50-80	080352	
P19-5S(+)	P19-2A (-)	180-220	080353	
TD201-3	E201	0.3 (BCC)	080354	
TD201-3	E202	40,000 - 55,000 (BCC)	080355	10 ma max
E213	P5-12	0.3 (BCC)	080356	
E215	P5-11	0.3 (BCC)	080357	
P19-5J	E220	0.3 (BCC)	080358	
P10-S	E219	0.3 (BCC)	080359	K2A-ON
P10-T	E221	0.3 (BCC)	080360	K2A-ON
K1A-11	P5-35	0.3 (BCC)	080361	K1B-ON
K1A-11	P19-2A	0.3 (BCC)	080362	K1A-ON
P7-M	P7-N	0.3	080363	
J15-D	J15-E	0.3	080364	
P10-S	E219	0.3 (BCC)	080365	K2B-ON
P10-T	E221	0.3 (BCC)	080366	K2B-ON
E219	E220	1.3-2.0 (BCC)	080367	
E220	E221	1.3-2.0 (BCC)	080368	
J3-H	CR209C	0.3 (BCC)	080369	
J3-H	CR208C	0.3 (BCC)	080370	
P5-14	CR212A	0.3 (BCC)	080371	
P19-1R	CR201A	0.3 (BCC)	080372	
P19-1S	CR204A	0.3 (BCC)	080373	
P19-3J	CR203C	0.3 (BCC)	080374	
C406 (+)	C405 (-)	0.3 (BCC)	080375	
C407 (+)	C408 (-)	0.3 (BCC)	080376	
P19-5S	K1A-19	0.3 (BCC)	080377	
P19-4U	K1A-19	0.3 (BCC)	080378	K1A-ON
P19-2A	K1B-9	0.3 (BCC)	080379	K1A-ON
E202	E217	0.3 (BCC)	080380	
E203	E217	0.3 (BCC)	080381	
P7-C	E322	0.3 (BCC	080382	
P5-36	P5-43	0.3	080383	
P19-2A(+)	P19-5R (-)	50-80	080384	K1A-ON
P19-5R(+)	P19-2A (-)	180-220	080385	K1A-ON
E203	E218	270-330 (BCC)	080386	
K1B-19	P19-4R	0.3 (BCC)	080387	K1B-ON
1J13-60	1J13-24		080388	See Note 1
1J13-57	1J13-24		080389	See Note 1
P19-5R	K1B-19	0.3	080390	
P19-2A	K1B-9	1 meg (min) (BCC)	080391	K1A-K1B-ON

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Table III (Cont'd)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)	Req'd
K1A-11	P19-2A	1 meg (min) (BCC)	080392	K1A-K1B-ON
K1B-9	P5-43	1 meg (min) (BCC)	080393	K1A-K1B-ON
P19-1K	1J12-D	0.3	080394	
P19-1K	1J13-66	0.3	080395	
P19-2C	1J12-E	0.3	080396	
P19-2C	1J3-67	0.3	080397	
1J1-B	1J12-S	0.3	080398	
1J1-N	J3-X	0.3	080399	
1J1-h	J3-h	0.3	080400	
1J1-c	J3-b	0.3	080401	
1J1-j	J3-q	0.3	080402	
1J12-B	J3-G	0.3	080403	
1J12-B	J3-K	0.3	080404	
1J12-B	1J13-26	0.3	080405	
1J12-T	1J13-54	0.3	080406	
1J12-T	1J13-53	0.3	080407	
1J12-T	1J13-52	0.3	080408	
1J12-h	1J12-g	0.3	080409	
1J12-h	J3-y	0.3	080410	
1J12-h	J3-x	0.3	080411	
1J12-h	1J13-23	0.3	080412	
1J12-A	J3-r	0.3	080413	
1J12-A	J3-t	0.3	080414	
J3-KK	J3-P	0.3	080415	
J3-k	J3-GG	0.3	080416	
1J12-T	1J13-45	0.3	080417	
1J12-T	J3-A	0.3	080418	
1J12-T	1J13-52	1 meg (min)	080419	K1B-ON
1J12-T	1J13-53	1 meg (min)	080420	K2A-ON
1J12-T	1J13-53	1 meg (min)	080421	K2B-ON
1J12-B	1J12-V	0.3	080422	
1J12-A	1J13-20	0.3	080423	
1J12-A	1J12-P	0.3	080424	
1J12-L	J3-T	0.3	080425	
1J12-L	J3-DD	0.3	080426	
1J12-L	1J12-C	0.3	080427	
1J12-L	1J13-41	0.3	080428	

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Note 1: For test codes 080388 and 080389; set Simpson 269 Meter to RX100 Scale. Connect positive lead of Simpson Meter to test points listed in FROM column. Connect negative lead of Simpson Meter to test point listed in the TO column. The Simpson Meter will slowly rise to a high resistance.
CAUTION: DO NOT REVERSE POLARITY.

3.10 Insulation resistance tests.

3.10.1 100 Megohm insulation resistance terminal board and harness assembly drawing 9317505. The insulation resistance between each circuit and all other circuits as indicated in Table IV shall not be less than 100 megohms when tested as specified in 4.4.2.1.

Table IV

From	To	From	To
P19-2R	All others & shell	P10-E	All others & shell
J3-n	"	J15-L	"
J3-D	"	J15-M	"
1J13-49	"	P4-A	"
P8-B	"	P4-B	"
J3-L	"	P4-E	"
J3-h	"	P4-F	"
1J13-67	"	P4-V	"
P4-C	"	P4-W	"
J3-S	"	P4-Z	"
J3-q	"	P4-a	"
J3-W	"	P4-b	"
J3-d	"	P4-CC	"
J3-X	"	P5-4	"
J3-j	"	P5-18	"
J3-b	"	P5-19	"
J3-s	"	P5-5	"
P19-5W	"	P5-28	"
J3-u	"	P5-8	"
J3-v	"	P5-29	"
1J13-20	"	P5-39	"
1J13-23	"	P5-40	"
1J13-66	"	P5-41	"
P4-AA	"	P5-45	"
J3-MM	"	P5-49	"
J3-HH	"	P5-50	"
J15-G	"	P5-51	"
		P5-52	
P6-H	"	1J13-45	"
P6-R	"	J3-C	"
P7-H	"	P6-M	"

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Table 1V (Cont'd)

From	To	From	To
P7-R	All others & shell	P5-22	All others & shell
P8-H	"	P5-23	"
P8-S	"	P5-25	"
P9-H	"	P5-46	"
P10-b	"	P10-a	"
1J13-6	"	P10-S	"
1J13-62	"	P10-T	"
1J13-63	"	P19-5J	"
1J2-M	"	P10-C	"
1J2-Y	"	P10-J	"
1J2-A	"	P10-M	"
1J2-J	"	P10-R	"
1J2-G	"	P10-U	"
	"	P10-V	"
1J2-W	"	P10-W	"
1J2-F	"	J3-F	"
1J13-47	"	J3-U	"
1J13-42	"	J3-Y	"
		J3-a	"
		J3-P	"
		J3-k	"
		1J13-26	"
		1J13-41	"

The following list of unused pins may be connected together and tested as one terminal:

P19-1M, 1N, 3D, 3L, 3M, 4F, 5A, 5B, 5C, 5D, 5E, 5F, 5G, 5H, 5N, 5P.

P4-L, P, R, d, e, f, g, h, k, m, n, p, q, r, s, t, u, v, x, y, z, EE, FF, GG, HH.

J3-B, N, V, Z, e, f, m, AA, PP.

P5-1, 2, 3, 10, 13, 17, 32, 34, 37, 38, 42, 44, 53, 54, 55.

P8-A, F, G, T.

P9-A, F, G, T.

P10-A, B, F, G, H, K, L, P, X, c.

J15-C, J, K, F, T, U.

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1J1-a, f, g, J.

1J2-T, Z.

1J12-M, a, b, d, W, j, K, U.

1J13-2, 4, 16, 17, 19, 21.

3.10.2 1 Megohm insulation resistance terminal board and harness assembly drawing 9317505. The insulation resistance between each circuit and all other circuits as indicated in Table V shall not be less than 1 megohm when tested as specified in 4.4.2.2.

Table V

From	To	From	To
1J13-13	All others & shell	1J16C	1J16S
P5-30	"	1J17C	1J17S
J3-z	"		
P5-31	"	P19-4U	P19-5S
J3-BB	"	P19-2R	J3-n
J15-D	"	J3-s	P19-5W
P4-T	"	P19-5R	P19-4R
P4-X	"	P10-S	P19-5J
P5-6	"	P10-T	P19-5J
P6-U	"	J3-C	P6-M
P4-G	"		
1J13-3	"		
P5-7	"		
P4-J	"		
P6-P	"		

3.11 Diode tests.--(BCC)

3.11.1 Reverse current leakage. The reverse current leakage of each circuit shall not exceed the values as shown in Table VI when tested as specified in 4.4.3.1.

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Table VI

Diode No. Under Test	+Terminal*	-Terminal*	Applied Reverse Direct Current Voltage ARVDC	Test Code (see 3.5)	Voltage Drop (max) Across Series Resistor at 200 ARVDC (see 4.4)
CR201 CR202	J3-M	J3-w	200 ± 10	102001	25
CR203 CR204	J3-M	J3-c	200 ± 10	102002	25
CR207 CR208	J3-H	P19-4W	200 ± 10	102003	25
CR209 CR210	J3-H	P19-4V	200 ± 10	102004	25
CR205	1J13-9	1J13-8	200 ± 10	102005	25
CR206	1J13-10	1J13-8	200 ± 10	102006	25
CR218	P4-T	P6-K	200 ± 10	102007	25
CR217	P4-T	P7-K	200 ± 10	102008	25
CR216	P4-K	P6-A	200 ± 10	102009	25
CR215	P4-K	P7-A	200 ± 10	102010	25

3.11.2 Forward voltage drop. The forward voltage drop of each circuit shall not exceed the maximum voltage values as shown in Table VII when tested as specified on 4.4.3.1.

*Polarity of measuring device.

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Table VII

Diode No. Under Test	+Terminal*	-Terminal*	Applied Forward Current (AMPS)	Test Code (see 3.5)	Max Forward Voltage Drop Volts (V)
CR201	J3-w	J3-M	5.0 ± 0.5	103001	3.5
CR202					
CR203	J3-c	J3-M	5.0 ± 0.5	103002	3.5
CR204					
CR207	P19-4W	J3-H	5.0 ± 0.5	103003	3.5
CR208					
CR209	P19-4V	J3-H	5.0 ± 0.5	103004	3.5
CR210					
CR205	1J13-8	1J13-9	5.0 ± 0.5	103005	3.5
CR206	1J13-8	1J13-10	5.0 ± 0.5	103006	3.5
CR215	P7-A	P4-K	1.0 ± 0.1	103007	2.0
CR216	P6-A	P4-K	1.0 ± 0.1	103008	2.0
CR217	P7-K	P4-T	1.0 ± 0.1	103009	2.0
CR218	P6-K	P4-T	1.0 ± 0.1	103010	2.0
CR211	1J13-11	1J13-7	5.0 ± 0.5	103011	3.5
CR212	1J13-12	1J13-7	5.0 ± 0.5	103012	3.5

3.12 Delay circuit test.--(BCC)

3.12.1 Time delay functioning.

3.12.1.1 Non-function. The delay unit shall not produce an output with a control input pulse width of 0.899 seconds +0.000 -0.100 seconds. Testing shall be as specfied in 4.4.4.3.1.

3.12.1.2 Function. The delay unit shall function and produce an output when a control input pulse width of 1.00 seconds ± 0.10 is applied. Testing shall be as specified in 4.4.4.3.2.

3.12.1.3 Output. The load current shall be 1.0 + 0.1 amperes steady state, or 7.0 ± 0.2 amperes for 10.0 ± 1.0 milliseconds. The voltage drop across the solid state switch shall not exceed 4 Volts with a current of 1.0 ± 0.1 ampere through the circuit.

3.13 Energy transfer circuit (ETC).--(BCC)

3.13.1 ETM output voltage. The output voltage of channel "A" ETM, and channel "B" ETM shall rise to a minimum of 50 volts in 10 microseconds (ms) maximum. At 20 ms the pulse voltage shall be no less than 25% of the peak voltage. Testing shall be as specified in 4.4.5.

3.14 Circuit resistance after assembly of control unit per dwg 9317506. The resistance of each circuit shall not

*Polarity of measuring device.

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exceed the maximum value listed in Table VIII or as shown otherwise when tested as specified in 4.4.1. Prior to the performance of this test, the item shall have successfully passed all the test requirements specified in 3.7, 3.8, 3.9, 3.10, 3.11, 3.12, 3.13, and 3.16.

Table VIII

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)
1J1-A	1J1-U	0.3	130001
1J1-B	1J13-47	0.3	130002
1J1-B	1J12-S	0.3	130003
1J1-D	1J1-E	0.3	130004
1J1-H	1J2-B	0.3	130005
1J1-H	1J13-42	0.3	130006
1J1-K	1J1-b	0.3	130007
1J1-K	1J13-63	0.3	130008
1J1-L	1J1-M	0.3	130009
1J1-L	1J13-62	0.3	130010
1J1-T	1J1-e	0.3	130011
1J1-X	1J1-W	0.3	130012
1J2-A	1J2-J	0.3	130013
1J2-C	1J2-R	0.3	130014
1J2-E	1J13-6	0.3	130015
1J2-E	1J2-Y	0.3	130016
1J2-F	1J2-W	0.3	130017
1J2-G	1J2-M	0.3	130018
1J2-S	1J13-46	0.3	130019
1J2-a	1J13-43	0.3	130020
1J2-b	P5-40	0.3	130021
1J2-c	P5-41	0.3	130022
1J13-24	P5-35	0.3	130023
1J13-24	P5-36	0.3	130024
1J13-24	P5-43	0.3	130025
1J13-57	1J13-24	See Note 1	130026
1J13-60	1J13-24	See Note 1	130027
1J13-61	1J13-64	0.3	130028
J3-A	1J13-55	0.3	130029
J3-A	1J13-54	0.3	130030
J3-A	1J13-53	0.3	130031
J3-A	1J13-45	0.3	130032
J3-A	1J13-52	0.3	130033
J3-A	1J12-T	0.3	130034
J3-D	1J2-N	0.3	130038
J3-D	1J2-P	0.3	130039
J3-E	P8-D	0.3	130040
J3-E	P8-B	0.3	130041

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Table VIII (Cont'd)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5
J3-E	P8-C	0.3	130042
J3-E	P8-E	0.3	130043
J3-E	P9-B	0.3	130044
J3-E	P9-C	0.3	130045
J3-E	P9-D	0.3	130046
J3-E	P9-E	0.3	130047
J3-E	1J13-25	0.3	130048
J3-F	J3-i	0.3	130049
J3-G	1J12-B	0.3	130050
J3-G	1J12-V	0.3	130052
J3-G	1J13-26	0.3	130054
J3-G	J3-K	0.3	130055
J3-H	1J13-13	0.3	130056
J3-H	J3-p	0.3	130057
J3-J	1J13-49	0.3	130058
J3-J	1J13-44	0.4	130059
J3-J	J15-G	0.4	130060
J3-L	P9-H	21-33	130063
J3-L	P9-J	21-33	130064
J3-L	P9-K	21-33	130065
J3-L	P9-L	21-33	130066
J3-L	P8-H	21-33	130067
J3-L	P8-J	21-33	130068
J3-L	P8-K	21-33	130069
J3-L	P8-L	21-33	130070
J3-M	1J13-3	0.3	130071
J3-M	P4-M	12-16	130072
J3-M	P4-N	12-16	130073
J3-M	P5-9	12-16	130074
J3-M	P5-16	12-16	130075
J3-M	P5-47	12-16	130076
J3-M	P5-48	12-16	130077
J3-M	P6-B	12-16	130078
J3-M	P6-C	12-16	130079
J3-M	P6-E	12-16	130080
J3-M	P6-G	12-16	130081
J3-M	P7-B	12-16	130082
J3-M	P7-C	12-16	130083
J3-M	P7-E	12-16	130084
J3-M	P7-G	12-16	130085
J3-M	1J13-24	12-16	130086
J3-R	P4-D	0.3	130089
J3-R	P4-C	0.3	130090
J3-S	1J1-R	0.3	130091
J3-T	1J12-C	0.3	130092

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Table VIII (Cont'd)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)
J3-T	1J12-L	0.3	130093
J3-T	1J13-41	0.3	130094
J3-T	J3-DD	0.3	130095
J3-U	J3-LL	0.3	130099
J3-W	1J1-P	0.3	130102
J3-X	1J1-N	0.3	130103
J3-Y	J3-JJ	0.3	130104
J3-a	J3-NN	0.3	130105
J3-b	1J1-c	0.3	130106
J3-c	P6-H	0.3	130107
J3-c	P6-J	0.3	130108
J3-g	J3-P	0.3	130112
J3-g	J3-KK	0.3	130113
J3-h	1J1-h	0.3	130114
J3-j	1J1-F	0.3	130115
J3-j	1J1-Z	0.3	130116
J3-k	J3-FF	0.3	130117
J3-k	J3-GG	0.3	130118
J3-q	1J1-j	0.3	130120
J3-r	J3-t	0.3	130122
J3-r	1J12-A	0.3	130125
J3-r	1J12-P	0.3	130126
J3-r	1J13-20	0.3	130127
J3-u	J3-HH	0.3	130128
J3-w	P7-H	0.3	130129
J3-w	P7-J	0.3	130130
J3-x	1J12-g	0.3	130131
J3-x	1J12-h	0.3	130132
J3-x	J3-y	0.3	130133
J3-MM	1J1-G	0.3	130134
J3-MM	1J1-Y	0.3	130135
J15-A	1J13-7	0.3	130136
J15-B	1J13-7	0.3	130137
J15-D	J15-E	0.3	130138
J15-D	1J13-48	0.3	130139
J15-G	1J13-44	0.3	130140
J15-H	1J13-40	0.3	130141
J15-P	1J13-51	0.3	130142
J15-L	J15-M	0.3	130143
J15-N	J15-P	0.3	130144
J15-N	1J13-51	0.3	130145
J15-R	1J13-35	0.3	130146
J15-S	1J13-39	0.3	130147
P4-A	1J2-H	0.3	130148

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Table VIII (Con'd)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)
P4-A	1J13-30	0.3	130149
P4-B	1J13-29	0.3	130150
P4-E	1J12-R	0.3	130151
P4-E	1J13-22	0.3	130152
P4-E	1J1-e	0.3	130153
P4-E	1J1-$\overline{T}$	0.3	130154
P4-F	1J1-A	0.3	130155
P4-F	1J1-U	0.3	130156
P4-F	1J12-e	0.3	130157
P4-G	1J13-$\overline{57}$	0.3	130158
P4-G	P4-H	0.3	130159
P4-G	P5-27	0.3	130160
P4-G	P5-33	0.3	130161
P4-J	P4-K	0.3	130163
P4-S	1J13-60	0.3	130164
P4-S	P5-26	0.3	130165
P4-T	P4-U	0.3	130166
P4-V	1J1-S	0.3	130167
P4-V	1J1-V	0.3	130168
P4-V	1J1-C	0.3	130169
P4-V	1J1-d	0.3	130170
P4-V	1J2-$\overline{L}$	0.3	130171
P4-V	1J2-V	0.3	130172
P4-V	P4-c	0.3	130173
P4-W	1J$\overline{13}$-28	0.3	130174
P4-X	1J2-Y	12-16	130175
P4-X	P5-11	8-12	130176
P4-X	P5-12	8-12	130177
P4-X	P5-20	12-16	130178
P4-X	P5-21	12-16	130179
P4-X	P5-28	12-16	130180
P4-X	P5-29	12-16	130181
P4-X	1J13-8	12-16	130182
P4-X	P4-Y	0.3	130183
P4-a	1J13-31	0.3	130184
P4-$\overline{a}$	P5-24	0.3	130185
P4-$\overline{AA}$	P4-BB	0.3	130186
P4-AA	P5-23	12-16	130187
P4-AA	P5-22	12-16	130188
P4-AA	P5-51	12-16	130189
P4-AA	P5-45	12-16	130190
P4-AA	1J13-15	12-16	130191
P4-CC	P4-DD	0.3	130192
P4-CC	1J12-c	0.3	130193

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Table VIII (Con'd)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)
P4-CC	1J12-N	0.3	130194
P4-CC	1J13-18	0.3	130195
P5-5	P8-N	0.3	130196
P5-5	P9-M	0.3	130197
P5-5	P5-7	0.9-2.5	130198
P5-5	P8-P	0.9-2.5	130199
P5-5	P9-U	0.9-2.5	130200
P5-5	1J13-9	0.9-2.5	130201
P5-6	1J13-10	0.3	130202
P5-6	P8-U	0.3	130203
P5-6	P9-P	0.3	130204
P5-6	P9-N	0.9-2.5	130205
P5-6	P8-M	0.9-2.5	130206
P5-6	P5-8	0.9-2.5	130207
P5-14	1J13-12	0.3	130208
P5-15	1J13-11	0.3	130209
P5-18	P5-19	0.3	130210
P5-18	P5-52	1.5-2.5	130211
P5-18	1J13-14	0.3	130212
P5-18	P5-46	1.5-2.5	130213
P5-25	P5-4	0.3	130214
P5-30	1J12-F	0.3	130215
P5-30	1J12-f	0.3	130216
P5-30	1J12-Y	0.3	130217
P5-30	1J12-Z	0.3	130218
P5-31	1J12-G	0.3	130219
P5-31	1J12-H	0.3	130220
P5-31	1J12-J	0.3	130221
P5-31	1J12-X	0.3	130222
P5-35	P5-43	0.3	130223
P5-35	P5-36	0.3	130224
P5-36	J15-M	12-16	130225
P5-39	1J13-33	0.3	130226
P5-39	P6-S	0.3	130227
P5-49	1J13-27	0.75-1.5	130228
P5-49	P5-50	0.75-1.5	130229
P6-D	1J13-11	270-330	130230
P6-D	1J13-12	270-330	130231
P6-F	P7-F	0.3	130232
P6-F	P7-D	0.3	130233
P6-F	P7-A	9-20	130234
P6-F	P7-L	9-20	130235
P6-F	P6-A	9-20	130236
P6-F	P6-L	9-20	130237

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Table VIII (Con'd)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)
P6-F	P6-D	0.3	130238
P7-K	P7-T	0.3	130239
P6-K	P6-T	0.3	130240
P6-M	P6-N	0.3	130241
P6-M	P7-M	27-35	130242
P6-M	P7-N	27-35	130243
P6-P	P7-P	0.3	130244
P6-P	1J2-K	0.3	130245
P6-P	1J2-U	0.3	130246
P6-R	P7-S	0.3	130247
P6-R	1J13-34	0.3	130248
P6-U	1J13-50	0.3	130249
P6-U	P7-U	0.3	130250
P6-U	1J2-D	0.3	130251
P6-U	1J2-X	0.3	130252
P7-R	P8-R	0.3	130253
P7-R	1J13-36	0.3	130254
P8-S	1J13-37	0.3	130255
P8-S	P9-R	0.3	130256
P9-S	1J13-38	0.3	130257
P9-S	1J13-40	0.4	130258
P9-S	J15-H	0.4	130259

Note: Set the Simpson 269 to RX100 scale. Connect the positive lead of the Simpson to the test points listed in the "FROM" column. Connect the negative lead of the Simpson to the test point listed in the "TO" column. The Simpson meter will slowly rise to a high resistance.
CAUTION: DO NOT REVERSE POLARITY.

3.15 Insulation resistance after assembly of control unit per dwg 9317506. The insulation resistance between each circuit and all other circuits, as indicated in Table IX, shall not be less than 100 megohms when tested as specified in 4.4.2. Prior to the performance of this test, the item shall have successfully passed all the test requirements specified in 3.7, 3.8, 3.9, 3.10, 3.11, 3.12, 3.13, and 3.16.

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Table IX

From	To	From	To
P4-A	All other listed circuits	1J13-23	All other listed circuits
P4-B	"	1J13-26	"
P4-C	"	1J13-41	"
P4-E	"	1J13-42	"
P4-F	"	1J13-45	"
P4-V	"	1J13-47	"
P4-W	"	1J13-56	"
P4-Z	"	1J13-61	"
P4-a	"	1J13-62	"
P4-CC	"	1J13-63	"
P5-4	"	1J13-66	"
P5-39	"	1J13-67	"
P5-40	"	J3-C	"
P5-41	"	J3-D	"
P6-M	"	J3-F	"
P6-P	"	J3-P	"
P6-R	"	J3-S	"
P6-U	"	J3-U	"
P7-R	"	P8-B	"
P9-S	"	P8-S	"
J15-G	"	J3-h	"
1J13-5	"	J3-j	"
1J13-20	"	J3-k	"
J3-W	"	J3-n	"
J3-X	"	J3-q	"
J3-Y	"	J3-s	"
J3-a	"	J3-v	"
J3-b	"	J3-MM	"
J3-d	"	1J2-A	"
		1J2-C	"
		1J2-F	"
		1J2-G	"

The following list of unused pins may be connected together and tested as one terminal:

J3-B, N, V, Z, e, f, m, AA, PP.
P4-L, P, R, d, e, f, g, h, k, m, n, p, q, r, s, t, u, v, w, x, y, z, EE, FF, GG, HH,
P5-1, 2, 3, 10, 13, 17, 32, 34, 37, 38, 42, 44, 53, 54, 55.
P8-A, F, G, T.
P9-A, F, G, T.

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J15-C, F, J, K, T, U.
1J1-J, a, f, g.
1J2-T, Z.
1J12-K, M, U, W, a, b, d, j.
1J13-2,4, 16, 17, 19, 21.

All connector shells shall be connected together as one terminal.

3.16 Relay functioning. --(BCC)
3.16.1 Pull-in voltage. The pull-in voltage shall be a maximum of 18.0 vdc. Testing shall be as specified in 4.4.6.1.

3.17 Temperature shock. The Control Unit shall withstand a temperature shock with temperatures ranging from -25°F to 140°F without evidence of cracking splitting separation of parts, or other damage. Testing shall be as specified in 4.4.7. At the completion of the final cycle, the Control Unit shall comply with the requirements of 3.14 and 3.15 when tested in accordance with the applicable test paragraphs of 4.4 respectively.

3.18 Vibration.
3.18.1 Transportation. The Control Unit shall withstand the vibration conditions specified in Table X along each of its three mutually perpendicular axes (see Figure 2). Frequency shall be cycled at a logarithmic sweep rate between 5-500-5 Hertz (Hz) in 15 minutes. Testing shall be as specified in 4.4.8. After this test, the Control Unit shall comply with the requirements of 3.14 and 3.15 when tested in accordance with the applicable paragraphs of 4.4 respectively.

Table X

Frequency (Hz)	Gravity Units (g's) or Double Amplitude (da) -25 + 5°F	140 + 5°F	Sweep rate
5-17-5	0.2" da ± 0.02	0.2" da ± 0.2	3 Frequency sweeps from 5-500-5 (15 minutes each sweep) for a total of 45 minutes per axis.
17-60-17	3.0 g's ± 0.03	3.0 g's ± 0.03	
60-80-60	10.0 g's ± 1.0g	10.0 g's ± 1.0g	
80-100-80	20.0 g's ± 2.0g	20.0 g's ± 2.0g	
100-500-100	5.0 g's ± 0.5g	5.0 g's ± 0.5g	

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3.18.2 Operation. The Control Unit shall be vibrated in each of three mutually perpendicular axes (see Figure 2), after being stabilized at -25°F and 140°F as applicable. The inputs shall be applied in the following sequence and conducted as specified in 4.4.

0.3 g^2/HZ power spectral density (19.0 root mean square (rms) g's min.) at frequencies ranging from 100 to 1000 Hz, attenuated 12 decibel (db)/octave so that the total frequency envelope is 50 to 2000 Hz for 10 seconds. Tolerance level: ± 3 db between frequencies of 100 and 1000 Hz and ± 6.0 db between frequencies of 1000-2000 Hz.

0.6 g^2/HZ power spectral density (8.5 rms g's min.) at frequencies ranging from 100 to 1000 Hz, attenuated 12 db/octabe so that the frequency envelope is 50 to 2000 Hz for one minute. Tolerance level: + 3 db between frequencies of 100-2000 Hz.

After completion of operational vibration test, the Control Unit shall be returned to ambient temperature and shall then meet the initial requirements of 3.14 and 3.15 when tested in accordance with the applicable test paragraphs of 4.4 respectively.

3.19 Mechanical shock. The Control Unit shall withstand a shock of 40 ± 4.0 gravity units (g's) along its longitudinal axis in the + x direction (see Figure 2) and 25 ± 2.5 g's both directions along each of its transverse and vertical axes (see Figure 2). The shocks shall be half-sine wave shape and shall be 11 ± 1.0 milliseconds (ms) duration along the longitudinal axis and 8 ± 0.8 ms duration along the transverse and vertical axes. Testing shall be as specified in 4.4.9. Upon completion of the test the Control Unit shall meet the requirements of 3.14 and 3.15 when tested in accordance with the applicable test paragraphs of 4.4 respectively.

3.20 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.21 Workmanship.

3.21.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.21.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.21.3 Burrs. Parts shall be free from burrs.

3.21.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

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3.21.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.21.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 4 complete assemblies (see dwg 9317506) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table XI. Tests shall be conducted in order specified in Table XI.

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Table XI

First article inspection and tests Control Unit No.					
Inspection or Tests	Requirement	1	2	3	4
1. Classification of Defects/ Characteristics	4.3.2.1 (except 4.3.2.1.4)	X	X	X	X
2. Circuit Resistance Diodes	3.11	X	X	X	X
3. Delay Unit Test	3.12	X	X	X	X
4. Relay Test	3.16	X	X	X	X
5. Energy Transfer Test	3.13	X	X	X	X
6. Classification of Defects/ Characteristics	4.3.2.1.4	X	X	X	X
7. Circuit Resistance	3.14	X	X	X	X
8. Insulation Resistance	3.15	X	X	X	X
9. Temperature Shock	3.17	X	X	X	X
10. Vibration at (160°F)	3.18	X	X	X	X
11. Vibration at (-25°F)	3.18	X	X	X	X
12. Mechanical Shock	3.19	X	X	X	X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation. The formation of an inspection lot and related functions shall be in accordance with Section 5, MIL-STD-105. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor defects/characteristics table (see Table XII and Table XIII), and inspected for all defects/characteristics in the applicable classification. Any defects/characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of defects/characteristics in each classification need not be followed as long as all defects/characteristics inspections are performed.

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Table XII

Major Defects/Characteristics

Lot Size	Sample Size
1-155	100%
156-200	155
201-250	175
251-300	190
301-350	200
351-400	210
401-450	220
451-500	225
501-550	230
551-600	235
601-650	240
651-700	243
701-750	247
751-800	250
801-900	254
901-1000	258
1001-1100	261
1101-1200	264
1201-1300	266
1301-1400	268
1401-1500	270
1501-1600	272

Table XIII

Minor Defects/Characteristics

Lot Size	Sample Size
1-25	100%
26-30	25
31-35	26
36-40	27
41-45	28
46-50	29
51-55	30
56-60	31
61-65	32
66-70	33
71-75	34
76-85	35
86-95	36
96-110	37

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Table XIII (Cont'd)

Minor Defects/Characteristics

Lot Size	Sample Size
111-130	38
131-156	39
157-190	40
191-240	41
241-340	42
341-520	43
521-1120	44
1121 and over	45

4.3.2.1 Classification of defects/characteristics (see 3.2). Critical, major and minor defects/characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Cover (see dwg 9236990) before assembly.

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Mislocation of holes (36)	Gage
102.	Incorrect intermediate size diameter clearance holes in flange (10)	Gage
103.	Incorrect largest diameter clearance holes in flange (2)	Gage
104.	Incorrect inside diameter	Gage
105.	Incorrect outside flange diameter	Gage
106.	Mislocation of cutout in mounting flange	Gage
107.	Evidence of poor workmanship (see 3.21)	Visual
108.	Incorrect external radius (max)	Gage
109.	Incorrect width of cutout in mounting flange	Gage
110.	Damage to protective finish exposing base metal	Visual
111.	Incorrect thickness of flange	Gage
112.	Incorrect flatness of flange	Gage
113.	Incorrect size of large cutout	Gage
114.	Part No., incorrect, illegible or missing	Visual
115.	Objective evidence relating to material missing or improper (see 3.1)	Visual

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Minor: None defined.

4.3.2.1.2 Plate, connector assembly (see dwg 9237019).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Mislocation of clinch nut mounting holes (11) (see dwg 9237017)	Gage
	102. Improper assembly of clinch nuts (11)	Visual/Manual
	103. Mislocation of connector mounting holes (6) (see dwg 9237017)	Gage
	104. Incorrect size of plate (see dwg 9237017)	Gage
	105. Evidence of poor workmanship (see 3.21)	Visual
	106. Incorrect thickness (see dwg 9237017)	Gage
	107. Damage to protective finish exposing base metal (see dwg 9237017)	Visual
	108. Marking illegible, missing or incorrect	Visual
	109. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.3 Mounting plate assembly (see dwg 9236980) inserts installed.

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Improper assembly of inserts (55 places)	Visual/Manual
	102. Evidence of poor workmanship (see 3.21)	Visual
	103. Objective evidence for casting missing or incomplete	Visual

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Categories	Defects/Characteristics	Method of Inspection

For the characteristics listed below, (see Plate mounting dwg 9236970).

104.	Mislocation of all tapped, clearance and alignment holes (86)	Gage
105.	Incorrect chamfer on outside diameter	Gage
106.	Incorrect width of cutout on mounting diameter	Gage
107.	Incorrect diameter of c'sink (Hole No. 1)	Gage
108.	Incorrect diameter of clearance hole on surface "C" (No. 7)	Gage
109.	Incorrect diameter of counterbore (No. 7)	Gage
110.	Incorrect depth of counter-bore (No. 7)	Gage
111.	Incorrect diameter of hole for connector alignment pins (12) (Nos. 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86)	Gage
112.	Incorrect inside diameter of "O" Ring groove (4) Nos. 32, 33, 35, 36)	Gage
113.	Incorrect width of "O" Ring groove (4) (Nos. 32, 33, 35, 36)	Gage
114.	Incorrect depth of "O" Ring groove (4) (Nos. 32, 33, 35, 36)	Gage
115.	Incorrect concentricity of "O" Ring groove (4) (No. 32, 33, 35, 36)	Gage
116.	Incorrect inside diameter of "O" Ring groove (No. 38)	Gage
117.	Incorrect width of "O" Ring groove (No. 38)	Gage
118.	Incorrect depth of "O" Ring groove (No. 38)	Gage
119.	Incorrect concentricity of "O" Ring groove (No. 38)	Gage
120.	Incorrect thickness of annular zone between outside diameter and 13.40 diameter	Gage
121.	Incorrect size of larger "D" shaped hole (NO. 39)	Gage
122.	Incorrect distance from con-nector mounting surface to datum surface "B" (6) (Nos. 32, 33, 34, 35, 36, 38)	Gage

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Categories	Defects/Characteristics	Method of Inspection
123.	Incorrect connector mounting flange thickness (8) (Nos.32, 33, 34, 35, 36, 37, 38, 39)	Gage
124.	Incorrect flatness of surface "B"	Gage
125.	Incorrect outside diameter	Gage
126.	Incorrect flatness of extended boss surfaces	Gage
127.	Incorrect parallelism of surface "C"	Gage
128.	Incorrect parallelism of mounting surface (boss side)	Gage
129.	Mislocation of outside diameter cutout	Gage
130.	Incorrect diameter of mounting clearance holes (6) Nos. 63, 64, 65, 66, 67, 68)	Gage
131.	Incorrect diameter of alignment pin holes (2) on mounting surface (Nos. 73 and 74)	Gage
132.	Incorrect chamfer of alignment pin holes (2) (Nos. 73 and 74)	Gage
133.	Improper surface finish	Visual
134.	Damage to protective finish exposing base metal	Visual
135.	Incorrect outside groove diameter	Gage
136.	Incorrect concentricity of outside groove diameter	Gage
137.	Incorrect width of groove	Gage
138.	Incorrect depth of groove	Gage
139.	Incorrect diameter (Hole No. 2)	Gage
140.	All bosses and standoffs covered with heat shrinkable tubing (see 9236980)	Visual
141.	All tubing neatly trimmed at ends of bosses and standoffs and no cracks, chips, or tears in tubing (see dwg 9236980)	Visual
142.	Improper assembly of spring pin in Hole No. 2 (see 9236980)	Gage
143.	Objective evidence relating to material missing or improper (see 3.1)	Visual

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Minor: None defined.

4.3.2.1.4 Control unit (see dwg 9317506).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Cover gasket improperly bonded to mounting plate assy (cover removed)	Visual/Manual
102.	Incorrect torque on connectors 14 places	Gage
103.	Connector plate gasket missing	Visual
104.	Incorrect torque on cover mounting screws (13 places)	Gage
105.	Incorrect torque on stand-offs (10 places) (cover removed)	Gage
106.	Incorrect torque for connector receptacle (cover removed)	Gage
107.	Incorrect torque for Gyro switch - 4 places (cover removed)	Gage
108.	Improper bonding of sealing disc	Visual/Manual
109.	Any component hardware or gasket, missing or loose or improperly assembled (cover removed)	Visual/Manual
110.	Evidence of poor workmanship (see 3.21)	Visual
111.	Serialization improper or missing (see 3.4)	Visual
112.	Objective evidence relating to material missing or improper (see 3.1)	Visual
113.	Incorrect torque for connector bracket to board (2 places)	Gage
Minor:	None defined.	

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4.3.2.1.5 Printed wiring board assembly (see dwg 9317502).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Improper or incorrect installation of squib switches (18)	Visual
102.	Improper or incorrect installation of resistors (18)	Visual
103.	Improper or incorrect installation of insulators (18)	Visual
104.	Cable strapped down securely	Visual/Manual
105.	Cable clear of mounting holes (2 places)	Visual
106.	Incorrect torque for strap mounting device (6) and connector to bracket (1)	Gage
107.	Incorrect or improper conformal coating	Visual
108.	Improper soldering (see 3.20)	Visual
109.	Marking improper, incorrect or illegible	Visual
110.	Evidence of poor workmanship (see 3.21)	Visual
*111.	Circuit resistance incorrect (see 3.7)	Gage
*112.	Insulation resistance incorrect (see 3.8)	Gage
113.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.6 Terminal board and harness assembly (see dwg 9317505).

Catergories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Any component mislocated, incorrect or improperly installed (see dwg 9317504)	Visual

*100% inspection is required.

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Categories	Defects/Characteristics	Method of Inspection
102.	Incorrect torque for installation of diodes (12) (see dwg 9317504)	Gage
103.	Incorrect torque for installation of relays (8) (see dwg 9317504)	Gage
104.	Incorrect torque for installation of delay unit (3 places)	Gage
105.	Incorrect torque for cabling mounting device (9) (see dwg 9317504)	Gage
106.	Improper soldering (see 3.20)	Visual
107.	Jumpers from top to bottom side of board not passing through grommets (see dwg 9317504)	Visual
108.	Any connector incorrect (16)	Visual
109.	Cable strapped down securely	Visual/Manual
110.	Incorrect or improper conformal coating	Visual
111.	Any component missing, loose or damaged	Visual/Manual
112.	Marking improper, illegible or incorrect	Visual
113.	Evidence of poor workmanship (see 3.21)	Visual
*114.	Circuit resistance incorrect (see 3.9)	Gage
*115.	Insulation resistance incorrect (see 3.10)	Gage
116.	Incorrect routing of wires	Visual
117.	Incorrect torque for installation of ETM (2) (see dwg 9317504)	Gage
118.	Objective evidence relating to material missing or improper (see 3.1)	Visual

Minor: None defined.

4.3.2.1.7 Preparation for delivery (see 5).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation, packing, and packaging not in accordance with MIL-STD-1169	Visual

*100% inspection is required.

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Categories	Defects/Characteristics	Method of Inspection
	102. Marking not in accordance with MIL-STD-1169	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.21)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Circuit resistance	Major
b. Insulation resistance	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Circuit resistance and insulation resistance. Each Control Unit in the lot shall be tested in accordance with 4.4.1 and 4.4.2. If any item fails to comply with 3.14 and 3.15, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major defects/characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). ELs for assemblies are listed on the Equipment Tabulation, ET-9317506. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor defects/characteristics and tests. Acceptance inspection equipment for minor defects/characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

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4.4.1 Circuit resistance.

4.4.1.1 Circuit resistance. The circuit resistance of each board (Printed & Terminal) and the assembled Control Unit shall be measured by utilizing a suitable circuit resistance tester. The measured resistance shall be specified in 3.7, 3.9 and 3.14. Current shall be limited to 10 milliamperes (ma) ± 0.10 ma, unless otherwise specified.

4.4.1.2 Circuit resistance. The circuit resistance shall be measured using a Simpson VOM Model 269. The measured resistance with the scale setting shall be as specified in 3.14.

4.4.2 Insulation resistance.

4.4.2.1 100 Megohm insulation resistance. An insulation resistance tester, capable of supplying 500 volts direct current (vdc) ± 50 (vdc), and with which the required resistance in megohms can be verified, shall be used to measure the insulation resistance. The current shall be limited to 10 ma (max). The measured resistance shall be as specified in 3.8, 3.10.1 and 3.15. The test voltage shall be applied for 5 seconds, minimum. The test time shall not exceed 60 seconds.

4.4.2.2 1 Megohm insulation resistance. An insulation resistance tester, capable of supplying 20 volts maximum direct current (vdc), and with which the required resistance in megohms can be verified, shall be used to measure the insulation resistance. The current shall be limited to 10 ma (max). The measured resistance shall be as specified in 3.8.1 and 3.10.2. The test voltage shall be applied for 5 seconds minimum. The test shall not exceed 60 seconds.

4.4.3 Diode tests.

4.4.3.1 Diode characteristics (reverse current leakage and forward voltage drop). Diodes shall be tested for forward voltage drop when the current specified in 3.11.2 is applied in the forward direction. Reverse current shall be determined by the voltage drop across a 1.0 ± 0.1 megohm resistor when the voltage is applied in the reverse direction as specified in 3.11.1. The duration of the applied signal in either the forward or the reverse direction shall not be more than 15 seconds.

4.4.4 Delay unit test.

4.4.4.1 Power input. The power input to the delay unit shall be as follows: Ebb +30 vdc ± 5 vdc; scr +30 vdc +5 vdc -4 vdc. Power will be applied simultaneously to pins 9, 8 and 1 and shall be a square type input voltage with a rise time of 3 ± 2 microseconds.

4.4.4.2 Control input voltage. The control input voltage to the delay unit shall be from +12 vdc to +35 vdc and shall be square type input voltage with a rise time of 5 ± 3 microseconds.

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4.4.4.3 Functioning.

4.4.4.3.1 Nonfunction. The delay unit shall be electrically connected as shown in Figure 1 and with control input pulse as specified in 3.12.1.1. The unit shall not produce any output. Power input shall be as specified in 4.4.4.1. Control input voltage as specified in 4.4.4.2.

4.4.4.3.2 Function. The delay unit shall be functioned in accordance with 3.12.1.2 when electrically connected as shown in Figure 1. Power shall be applied as specified in 4.4.4.1.

4.4.5 ETM output voltage. An external load of 1.0 ± 0.1 ohm, 1 watt non-inductive resistor connected in series with a 0.5 ± 0.02 ohm, 1 watt non-inductive resistor shall be connected across the output terminal of each channel ("A & B") as shown in Figure 3. Momentarily depress SW-1 and record output voltage across external 1.0 ohm load for channel "A". Record for channel "B" by momentarily depressing SW-2. This method, in conjunction with Figure 3, is a suggested method. Any other method which is equivalent will be acceptable, contingent on Government approval.

4.4.6 Relay functioning.

4.4.6.1 Pull-in voltage. A suitable dc voltage measuring device shall be connected to the coil terminals and shall be used to measure the pull-in voltage which will consistently operate the relay. Three consecutive pull-ins shall be considered consistent performance. See Figure 4 for coil and contact terminals.

4.4.7 Temperature shock. The Control Unit shall be tested in accordance with MIL-STD-810, Method 503. At the completion of tests, the Control Unit shall meet the requirements of 3.17.

4.4.8 Vibration. The Control Unit shall be mounted securely to a suitable fixture and vibrated at the levels and conditions specified in 3.18.1. After undergoing the operational vibration, the Control Unit shall meet the post-test requirements of 3.18.

4.4.9 Mechanical shock. The Control Unit assembly shall be mounted to a suitable fixture and shocked at the levels and conditions specified in 3.19. After undergoing the mechanical shock, the Control Unit shall meet the post-test requirements of 3.19.

4.4.10 Operational vibration. The Control Unit shall be mounted to a suitable fixture and vibrated at the levels and conditions specified in 3.18.2. After undergoing vibration, the Control Unit shall meet the post-test requirements of 3.18.2.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The Control Unit shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.1.2 Level C. The Control Unit shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.2 Packing.

5.2.1 Levels A and C. The Control Unit shall be packed in accordance with the applicable instructions contained in MIL-STD-1169.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 3.5). Data specified in 3.5, 3.7, 3.9, 3.11 and 3.14 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Dover, New Jersey 07801, ATTN: DRDAR-QAN-I.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min do not exceed the nominal temperature plus tolerance specified.

6.5 Drawings and publications. Drawings and publications listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings and publications prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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Preparing activity:
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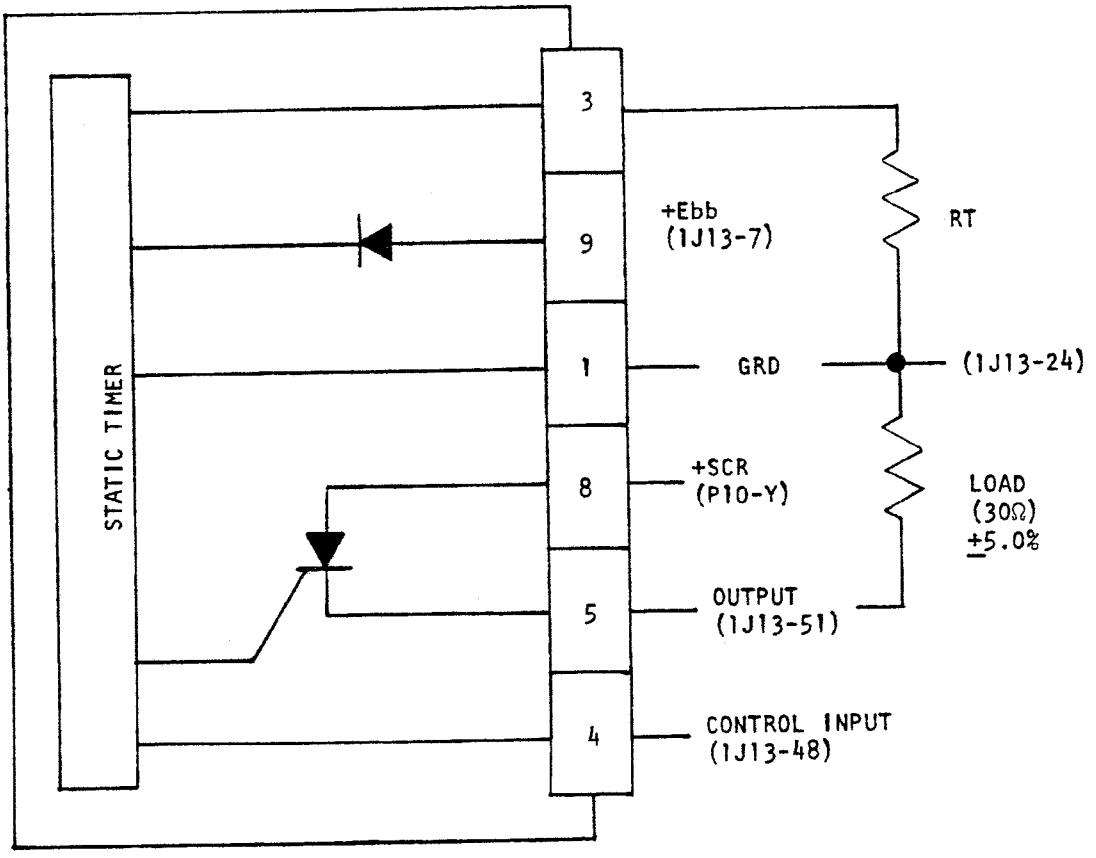


Figure 1. Delay Unit

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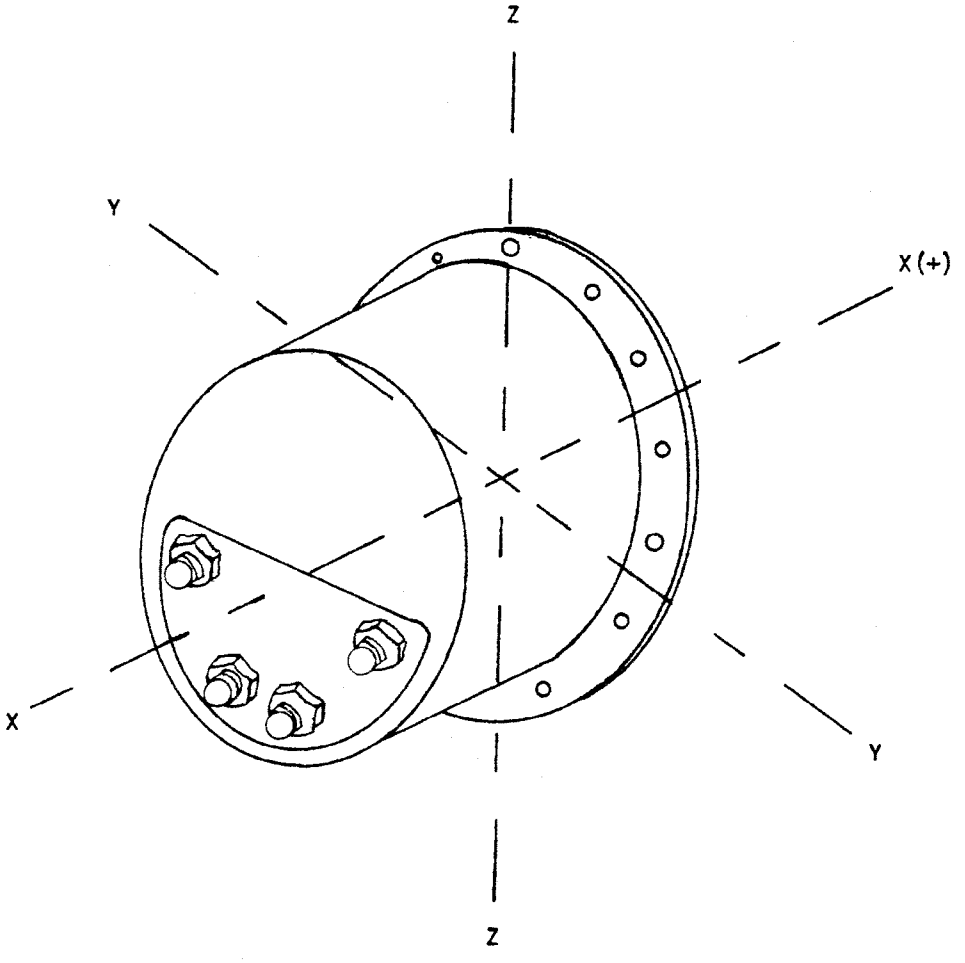


Figure 2. Control Unit

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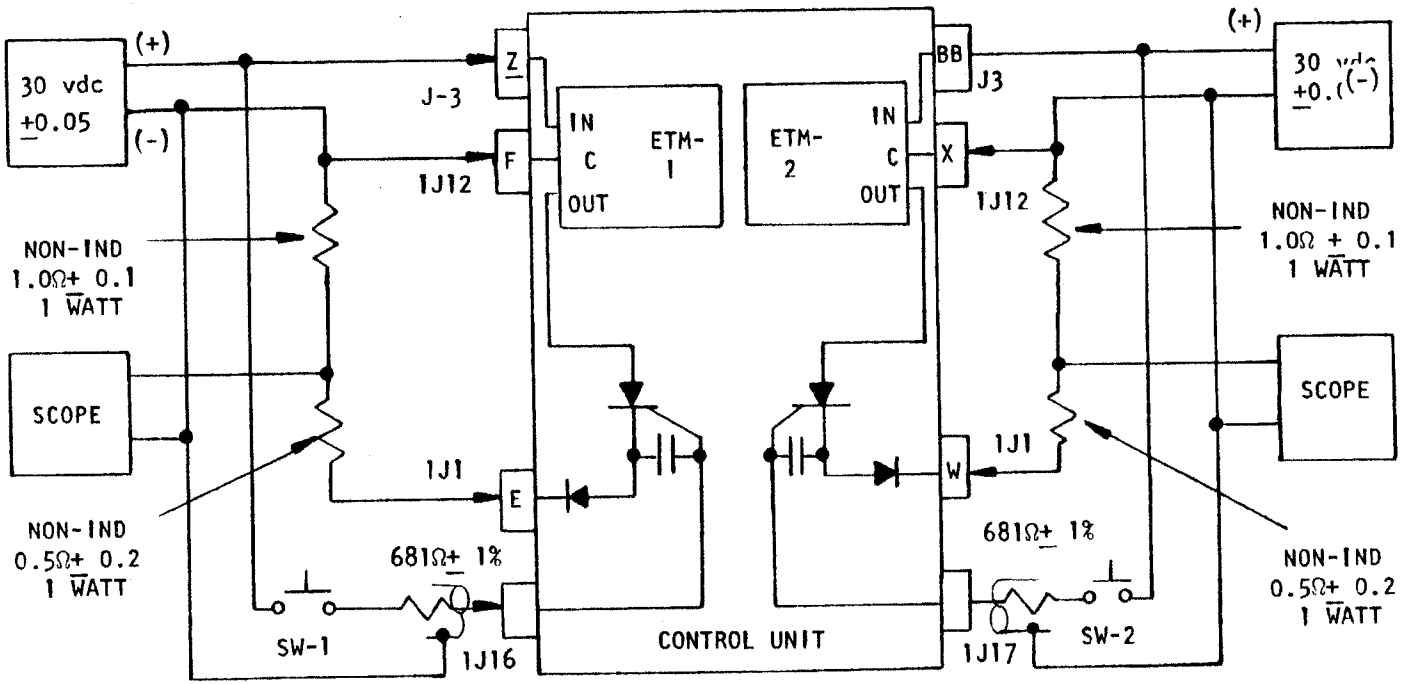
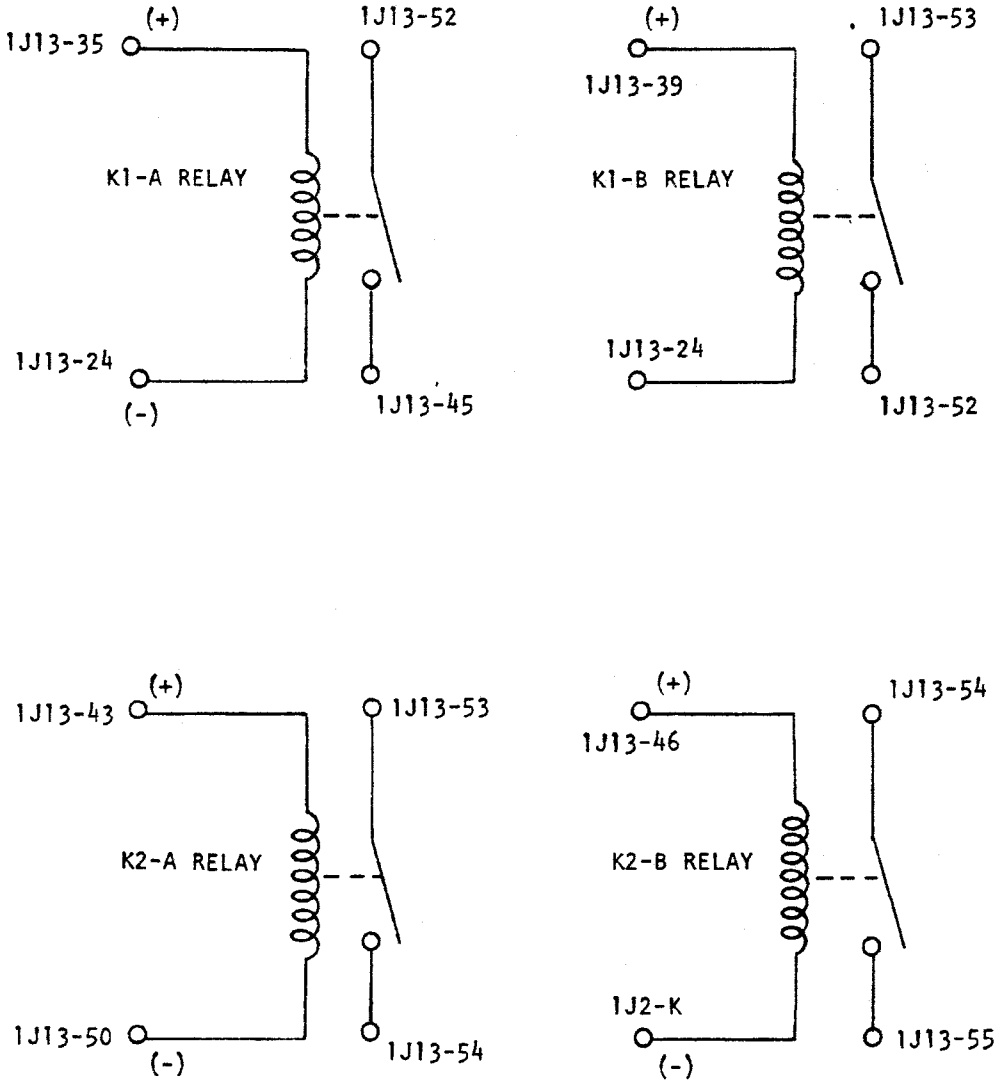


Figure 3. Energy Transfer Module Test Setup

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NOTE: RELAYS SHOWN ARE IN THE ENERGIZED POSITION.

Figure 4. K1, K2 Relays

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-63177A Control Unit: Lance AK (M238E1)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO

DATE

DD FORM 1426 1 OCT 76

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

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